

The Complex Orthogonal Measure and the Spectral Representation of the Hardy Z -Function

Abstract

Let θ be the Riemann–Siegel theta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z -function, and $H(x) := Z(\theta^{-1}(x))/\sqrt{\theta'(\theta^{-1}(x))}$ the warped Hardy signal. Given as input the companion results

- Theorem A of *Three Missing Theorems*: $C(\eta) := \lim_{X \rightarrow \infty} C_X(\eta) = 2 \cos |\eta|$,
- the Bochner factorization $C(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\nu(\xi)$ with $\nu = \delta_{-1} + \delta_{+1}$ from *Existence of the Spectral Measure in the Limit*,

we construct, without any probabilistic hypothesis added beyond these two inputs, the complex orthogonal measure Φ of H and prove the spectral representation

$$H(x) = \int_{\mathbb{R}} e^{i\xi x} d\Phi(\xi)$$

as an identity in the Hilbert space generated by the shifts of H under the time-average inner product. The measure Φ is a Hilbert-space-valued orthogonally-scattered measure on the Borel σ -algebra of $\mathbb{R}$, supported on $\{-1, +1\}$, with intensity $\langle \Phi(A), \Phi(B) \rangle = \nu(A \cap B)$, and the two atoms $\Phi(\{-1\}), \Phi(\{+1\})$ are orthonormal in that Hilbert space. The corresponding spectral representation of the Hardy Z -function is

$$Z(t) = \sqrt{\theta'(t)} \int_{\mathbb{R}} e^{i\xi\theta(t)} d\Phi(\xi) \quad \text{in the same Hilbert space.}$$

1 Inputs and notation

Fix $T_0 \geq 200$ and set $X_0 := \theta(T_0)$. The Riemann–Siegel theta function

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$$

is a C^∞ -diffeomorphism of $[T_0, \infty)$ onto $[X_0, \infty)$ with $\theta' > 0$ there. The Hardy Z -function is $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$, which is real-valued on $\mathbb{R}$. The warped Hardy signal is

$$H(x) := \frac{Z(\theta^{-1}(x))}{\sqrt{\theta'(\theta^{-1}(x))}}, \quad x \in [X_0, \infty),$$

and H is a locally integrable (in fact continuous) real-valued function on $[X_0, \infty)$. For $\eta \in \mathbb{R}$ and $X > X_0 + |\eta|$ define the one-sided time-averaged covariance

$$C_X(\eta) := \frac{1}{X - X_0} \int_{X_0}^{X-|\eta|} H(x) \overline{H(x + |\eta|)} dx. \quad (1)$$

Since H is real, $C_X(\eta) = C_X(-\eta)$.

Inputs cited.

Input 1 (Theorem A). For every $\eta \in \mathbb{R}$,

$$C(\eta) := \lim_{X \rightarrow \infty} C_X(\eta) = 2 \cos |\eta|,$$

with an explicit quantitative rate $|C_X(\eta) - C(\eta)| \leq 8\pi(1 + |\eta|) \log^3(X/X_0)/(X - X_0) + 0.349/\sqrt{T_0}$.

Input 2 (Bochner measure). There is a unique finite positive Borel measure ν on $\mathbb{R}$ with $C(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\nu(\xi)$ for all $\eta \in \mathbb{R}$, and

$$\nu = \delta_{-1} + \delta_{+1}, \quad \nu(\mathbb{R}) = 2.$$

Shift operator. For $s \in \mathbb{R}$ and a function f on $[X_0, \infty)$, we write $(T_s f)(x) := f(x + s)$, defined on $[X_0 - s, \infty) \cap [X_0, \infty)$; since our inner product below integrates over an interval that eventually contains any bounded translation set, all finite shifts are well-defined members of the Hilbert space constructed.

2 The time-average Hilbert space

We construct a complex Hilbert space $\mathcal{H}$ in which the shifts $T_s H$, $s \in \mathbb{R}$, live and in which Input 1 reads as a positive-definite kernel.

Definition 1 (Time-average sesquilinear form on trigonometric combinations). Let $\mathcal{V}$ be the complex vector space spanned by the formal symbols $T_s H$, $s \in \mathbb{R}$: its elements are finite formal linear combinations $f = \sum_{j=1}^n a_j T_{s_j} H$ with $a_j \in \mathbb{C}$, $s_j \in \mathbb{R}$, modulo the obvious linearity relations (so $T_s H + T_s H = 2T_s H$ etc.). Every such f determines an actual function on $[X_0 + \max_j |s_j|^{-}, \infty)$ where $|s_j|^{-} := \max(-s_j, 0)$ is the amount of leftward shift, by

$$f(x) = \sum_{j=1}^n a_j H(x + s_j), \quad x \geq X_0 + \max_j |s_j|^{-}.$$

For $f = \sum_j a_j T_{s_j} H$ and $g = \sum_k b_k T_{u_k} H$ in $\mathcal{V}$, set

$$\langle f, g \rangle := \sum_{j,k} a_j \overline{b_k} C(u_k - s_j). \quad (2)$$

Lemma 2 (The form is well-defined, sesquilinear, and equal to a time average). *For every pair of formal symbols $T_s H, T_u H$ with $s, u \geq 0$,*

$$C(u - s) = \lim_{X \rightarrow \infty} \frac{1}{X - X_0} \int_{X_0}^{X - \max(s, u)} H(x + s) \overline{H(x + u)} dx, \quad (3)$$

and the inner product of two elements $f = \sum_j a_j T_{s_j} H$, $g = \sum_k b_k T_{u_k} H$ with all $s_j, u_k \geq 0$ is

$$\langle f, g \rangle = \lim_{X \rightarrow \infty} \frac{1}{X - X_0} \int_{X_0 + M}^X f(x) \overline{g(x)} dx \quad (4)$$

where $M := \max_j s_j \vee \max_k u_k$. The form (2) is sesquilinear and Hermitian, and the value (2) does not depend on the representation of f and g .

Proof. For (3): by the substitution $y = x + s$ (with $s \geq 0$),

$$\int_{X_0}^{X-\max(s,u)} H(x+s)\overline{H(x+u)} dx = \int_{X_0+s}^{X-\max(s,u)+s} H(y)\overline{H(y+(u-s))} dy.$$

Set $\eta := u - s$ (whose sign is arbitrary); WLOG $\eta \geq 0$ (else use $C(-\eta) = C(\eta)$ and swap $s \leftrightarrow u$, $f \leftrightarrow g$). Then $\max(s, u) = u = s + \eta$ and the integral becomes $\int_{X_0+s}^{X-\eta+s} H(y)\overline{H(y+\eta)} dy$. Comparing with $(X - X_0)C_X(\eta) = \int_{X_0}^{X-\eta} H(y)\overline{H(y+\eta)} dy$, the two integration intervals differ by two sets each of length s (the left window $[X_0, X_0 + s]$ removed, the right window $[X - \eta, X - \eta + s]$ added). Denote the integral of $H(y)\overline{H(y+\eta)}$ over such a window J of length s by E_J . Cauchy–Schwarz gives

$$|E_J| \leq \left(\int_J |H(y)|^2 dy \right)^{1/2} \left(\int_{J+\eta} |H(y)|^2 dy \right)^{1/2}.$$

Input 1 at $\eta = 0$ says

$$\int_{X_0}^X |H(y)|^2 dy = (X - X_0)(2 + o(1)) \quad (X \rightarrow \infty), \quad (5)$$

and consequently for any interval J of fixed length s and right endpoint at distance $\leq s + \eta$ from X ,

$$\int_J |H(y)|^2 dy \leq \int_{X_0}^{X+s+\eta} |H|^2 - \int_{X_0}^{X-s-\eta} |H|^2 = 2(2s + 2\eta) + o(X) = O(s + \eta) + o(X).$$

For an interval $J \subseteq [X_0, X_0 + 2s]$ of length s , $\int_J |H|^2 dy$ is bounded by the finite constant $\int_{X_0}^{X_0+2s} |H|^2 dy$, which is a fixed number (continuity of H). In either case $|E_J| \leq O(s + \eta) + o(X)$, and this divided by $X - X_0$ tends to 0. Summing over the two endpoint windows,

$$\left| \int_{X_0+s}^{X-\eta+s} H(y)\overline{H(y+\eta)} dy - \int_{X_0}^{X-\eta} H(y)\overline{H(y+\eta)} dy \right| = O(s + \eta) + o(X).$$

Dividing by $X - X_0$ and using $\lim_X C_X(\eta) = C(\eta)$ (Input 1),

$$\lim_{X \rightarrow \infty} \frac{1}{X - X_0} \int_{X_0}^{X-\max(s,u)} H(x+s)\overline{H(x+u)} dx = C(u - s).$$

For (4): expand $f\bar{g} = \sum_{j,k} a_j \bar{b}_k H(x + s_j) \overline{H(x + u_k)}$, integrate from $X_0 + M$ to X , divide by $X - X_0$, and take $X \rightarrow \infty$. The limit of each summand is $a_j \bar{b}_k \cdot C(u_k - s_j)$ by (3) applied with X replaced by $X - M$ (the M -offset is absorbed into the $o(X)$ error). Summing gives (2).

Sesquilinearity on both slots is immediate from (2). Hermitian symmetry $\langle f, g \rangle = \overline{\langle g, f \rangle}$ amounts to $C(-\eta) = \overline{C(\eta)}$, which holds because $C(\eta) = 2 \cos |\eta|$ is real.

Independence of the representation of f and g : by (4) the value $\langle f, g \rangle$ depends only on the functions $x \mapsto f(x), g(x)$ on any right half-line $[X_0 + M, \infty)$, not on the formal representation.

For symbols $T_s H$ with $s < 0$, extend the inner product to $\mathcal{V}$ by the formula (2), which is defined for any $s_j, u_k \in \mathbb{R}$ because C is. Sesquilinearity and Hermitian symmetry are again immediate. The time-average formula (4) then extends to arbitrary real shifts, integrating from $X_0 + \max_j(-s_j, -u_k, 0)$: the proof is identical, with s replaced by $|s|$ in the endpoint-window bound. $\square$

Lemma 3 (Positive-semidefiniteness). *The form $\langle \cdot, \cdot \rangle$ on $\mathcal{V}$ is positive semidefinite: $\langle f, f \rangle \geq 0$ for every $f \in \mathcal{V}$.*

Proof. By Input 2, $C(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\nu(\xi)$. For $f = \sum_j a_j T_{s_j} H \in \mathcal{V}$,

$$\langle f, f \rangle = \sum_{j,k} a_j \overline{a_k} C(s_k - s_j) = \sum_{j,k} a_j \overline{a_k} \int_{\mathbb{R}} e^{i(s_k - s_j)\xi} d\nu(\xi) = \int_{\mathbb{R}} \left| \sum_j a_j e^{-is_j\xi} \right|^2 d\nu(\xi) \geq 0.$$

□

Definition 4 (The Hilbert space $\mathcal{H}$). Let $\mathcal{N} := \{f \in \mathcal{V} : \langle f, f \rangle = 0\}$; by Cauchy–Schwarz for the positive semidefinite form, $\mathcal{N}$ is a linear subspace of $\mathcal{V}$. Equip the quotient $\mathcal{V}/\mathcal{N}$ with the induced inner product $\langle \cdot, \cdot \rangle$. Let $\mathcal{H}$ be the Hilbert-space completion of $\mathcal{V}/\mathcal{N}$.

By construction, the map $\mathcal{V} \ni f \mapsto [f] \in \mathcal{V}/\mathcal{N} \subseteq \mathcal{H}$ is linear and length-preserving for $\langle \cdot, \cdot \rangle$. We denote the class $[T_s H] \in \mathcal{H}$ simply by H_s and write $H_0 = H$ when we mean $[H]$. Then

$$\langle H_s, H_t \rangle_{\mathcal{H}} = C(s - t) = 2 \cos(s - t), \quad s, t \in \mathbb{R}. \quad (6)$$

Lemma 5 (Strong continuity of the shift group). *The map $\mathbb{R} \ni s \mapsto H_s \in \mathcal{H}$ is continuous; more precisely,*

$$\|H_s - H_t\|_{\mathcal{H}}^2 = 2C(0) - 2\operatorname{Re} C(s - t) = 4 - 4\cos(s - t) = 4\sin^2 \frac{s-t}{2},$$

$$\text{so } \|H_s - H_t\|_{\mathcal{H}} = 2 \left| \sin \frac{s-t}{2} \right|.$$

Proof. Expand $\|H_s - H_t\|_{\mathcal{H}}^2 = \langle H_s - H_t, H_s - H_t \rangle$ using (6) and $C(0) = 2$, $C(s - t) = 2\cos(s - t)$. □

3 The Kolmogorov isometry and the complex orthogonal measure

Definition 6 (The closed span). Let $\mathcal{H}_H := \overline{\operatorname{span}}\{H_s : s \in \mathbb{R}\} \subseteq \mathcal{H}$, the closed linear span in $\mathcal{H}$ of the shifts of H .

Theorem 7 (Cramér/Kolmogorov isometry). *There is a unique isometric linear isomorphism*

$$V : \mathcal{H}_H \longrightarrow L^2(\mathbb{R}, \nu)$$

such that $V(H_s)(\xi) = e^{i\xi s}$ for every $s \in \mathbb{R}$.

Proof. Define $V_0 : \operatorname{span}\{H_s : s \in \mathbb{R}\} \rightarrow L^2(\nu)$ by linear extension of $H_s \mapsto e^{i\xi s}$. This is well-defined: for $f = \sum_j a_j H_{s_j} \in \operatorname{span}\{H_s\}$,

$$\|f\|_{\mathcal{H}}^2 = \sum_{j,k} a_j \overline{a_k} C(s_k - s_j) = \int_{\mathbb{R}} \left| \sum_j a_j e^{is_j\xi} \right|^2 d\nu(\xi) = \left\| \sum_j a_j e^{is_j\xi} \right\|_{L^2(\nu)}^2,$$

so V_0 is well-defined and isometric (in particular, whenever $\sum_j a_j H_{s_j} = 0$ in $\mathcal{H}$, also $\sum_j a_j e^{i\xi s_j} = 0$ in $L^2(\nu)$, and conversely). By the bounded linear extension theorem, V_0 extends uniquely to an isometry $V : \mathcal{H}_H \rightarrow L^2(\nu)$.

Surjectivity. Since $\nu = \delta_{-1} + \delta_{+1}$ is supported on two points, $L^2(\nu) \simeq \mathbb{C}^2$ under $f \mapsto (f(-1), f(+1))$. The two vectors $V(H_0) = (1, 1)$ and $V(H_\pi) = (e^{-i\pi}, e^{i\pi}) = (-1, -1)$ are parallel, so we need another point. Take $V(H_{\pi/2}) = (e^{-i\pi/2}, e^{i\pi/2}) = (-i, i)$. The vectors $(1, 1)$ and $(-i, i)$ are linearly independent in $\mathbb{C}^2$, so the image $V(\operatorname{span}\{H_s\})$ contains a two-dimensional subspace of $\mathbb{C}^2$, hence equals $\mathbb{C}^2 = L^2(\nu)$. Since V is isometric with dense range in the complete space $L^2(\nu)$, it is surjective. □

Definition 8 (The complex orthogonal measure Φ). For every Borel set $A \subseteq \mathbb{R}$, define

$$\Phi(A) := V^{-1}(\mathbf{1}_A) \in \mathcal{H}_H \subseteq \mathcal{H}.$$

By construction,

$$\|\Phi(A)\|_{\mathcal{H}}^2 = \|\mathbf{1}_A\|_{L^2(\nu)}^2 = \nu(A), \quad (7)$$

so $\Phi(A) \in \mathcal{H}$ has finite norm for every Borel set A and $\Phi(A) = 0$ whenever $\nu(A) = 0$.

Theorem 9 (Properties of Φ). *The map $\Phi: \mathcal{B}(\mathbb{R}) \rightarrow \mathcal{H}$ defined by $\Phi(A) := V^{-1}(\mathbf{1}_A)$ satisfies*

- (i) $\langle \Phi(A), \Phi(B) \rangle_{\mathcal{H}} = \nu(A \cap B)$ for all $A, B \in \mathcal{B}(\mathbb{R})$ (orthogonal intensity);
- (ii) Φ is finitely additive: $\Phi(A \sqcup B) = \Phi(A) + \Phi(B)$;
- (iii) Φ is countably additive in the norm of $\mathcal{H}$: if $A = \bigsqcup_{n \geq 1} A_n$ is a Borel partition, then $\sum_{n=1}^N \Phi(A_n) \rightarrow \Phi(A)$ in $\mathcal{H}$ as $N \rightarrow \infty$;
- (iv) $\Phi(\mathbb{R}) = \Phi(\{-1\}) + \Phi(\{+1\})$, with $\|\Phi(\{-1\})\| = \|\Phi(\{+1\})\| = 1$ and $\langle \Phi(\{-1\}), \Phi(\{+1\}) \rangle_{\mathcal{H}} = 0$.

Proof. (i) By the polarisation of (7),

$$\langle \Phi(A), \Phi(B) \rangle_{\mathcal{H}} = \langle V^{-1}\mathbf{1}_A, V^{-1}\mathbf{1}_B \rangle_{\mathcal{H}} = \langle \mathbf{1}_A, \mathbf{1}_B \rangle_{L^2(\nu)} = \int_{\mathbb{R}} \mathbf{1}_A \mathbf{1}_B d\nu = \nu(A \cap B).$$

(ii) For disjoint Borel A, B , $\mathbf{1}_{A \sqcup B} = \mathbf{1}_A + \mathbf{1}_B$, so applying V^{-1} (which is linear) gives $\Phi(A \sqcup B) = \Phi(A) + \Phi(B)$.

(iii) Set $S_N := \sum_{n=1}^N \mathbf{1}_{A_n} = \mathbf{1}_{A_1 \sqcup \dots \sqcup A_N} \uparrow \mathbf{1}_A$ monotonically. Since $|\mathbf{1}_A - S_N| \leq 1 \in L^2(\nu)$ (a constant function is in $L^2(\nu)$ because ν is finite) and $\mathbf{1}_A - S_N \rightarrow 0$ pointwise, dominated convergence on $L^2(\nu)$ gives $\|\mathbf{1}_A - S_N\|_{L^2(\nu)} \rightarrow 0$. Applying the isometry V^{-1} ,

$$\left\| \Phi(A) - \sum_{n=1}^N \Phi(A_n) \right\|_{\mathcal{H}} = \|\mathbf{1}_A - S_N\|_{L^2(\nu)} \xrightarrow{N \rightarrow \infty} 0.$$

(iv) Since $\nu = \delta_{-1} + \delta_{+1}$ is supported on $\{-1, +1\}$ and $\mathbb{R} = \{-1\} \sqcup \{+1\} \sqcup (\mathbb{R} \setminus \{-1, +1\})$ with $\nu(\mathbb{R} \setminus \{-1, +1\}) = 0$, part (iii) together with (7) (which forces $\Phi(\mathbb{R} \setminus \{-1, +1\}) = 0$) yields $\Phi(\mathbb{R}) = \Phi(\{-1\}) + \Phi(\{+1\})$. The norms $\|\Phi(\{\pm 1\})\|_{\mathcal{H}}^2 = \nu(\{\pm 1\}) = 1$ give unit length, and orthogonality $\langle \Phi(\{-1\}), \Phi(\{+1\}) \rangle = \nu(\{-1\} \cap \{+1\}) = \nu(\emptyset) = 0$ comes from (i). $\square$

4 The spectral representation of H

Definition 10 (Stochastic integral against Φ). For $g \in L^2(\nu)$, define

$$I(g) := V^{-1}(g) \in \mathcal{H}_H.$$

We write this equivalently as

$$I(g) = \int_{\mathbb{R}} g(\xi) d\Phi(\xi).$$

The notation is justified by: for any simple function $g = \sum_j c_j \mathbf{1}_{A_j} \in L^2(\nu)$ with disjoint $A_j \in \mathcal{B}(\mathbb{R})$, linearity of V^{-1} gives

$$I(g) = \sum_j c_j \Phi(A_j),$$

which is the classical definition of the integral of a simple function against a vector-valued orthogonally-scattered measure; and simple functions are dense in $L^2(\nu)$, so I as defined extends the simple-function integral by continuity, agreeing with the general definition of a Dunford–Schwartz/Riesz integral against an orthogonal measure.

Theorem 11 (Spectral representation of H). *For every $x \in \mathbb{R}$,*

$$H_x = \int_{\mathbb{R}} e^{i\xi x} d\Phi(\xi) \quad \text{in } \mathcal{H}. \quad (8)$$

In particular, the function $\xi \mapsto e^{i\xi x}$ belongs to $L^2(\nu)$ for each $x \in \mathbb{R}$ (it is bounded, ν is finite), and the identity (8) holds in the norm of $\mathcal{H}$.

Proof. By definition of V (Theorem 7), $V(H_x)(\xi) = e^{i\xi x}$, i.e. $V(H_x) = e^{i\xi x}$ as an element of $L^2(\nu)$. Applying V^{-1} to both sides and using Definition 10,

$$H_x = V^{-1}(e^{i\xi x}) = I(e^{i\xi x}) = \int_{\mathbb{R}} e^{i\xi x} d\Phi(\xi).$$

□

Corollary 12 (Explicit two-atom form). *Writing $\Phi_{\pm} := \Phi(\{\pm 1\})$, the spectral representation (8) reduces to*

$$H_x = e^{-ix} \Phi_- + e^{+ix} \Phi_+ \quad \text{in } \mathcal{H}, \quad (9)$$

where $\{\Phi_-, \Phi_+\}$ is orthonormal in $\mathcal{H}$.

Proof. Since $\nu = \delta_{-1} + \delta_{+1}$, for any $g \in L^2(\nu)$ the integral $\int g d\nu = g(-1) + g(+1)$, and the corresponding simple function representation of g in $L^2(\nu)$ is $g = g(-1)\mathbf{1}_{\{-1\}} + g(+1)\mathbf{1}_{\{+1\}}$ (they agree ν -a.e.). Applying V^{-1} ,

$$I(g) = g(-1)\Phi_- + g(+1)\Phi_+.$$

Taking $g(\xi) = e^{i\xi x}$ gives (9). Orthonormality of $\{\Phi_-, \Phi_+\}$ is Theorem 9(iv). □

Corollary 13 (Reality constraint). *For every $s \in \mathbb{R}$,*

$$\langle H_s, \Phi_- \rangle_{\mathcal{H}} = e^{-is}, \quad \langle H_s, \Phi_+ \rangle_{\mathcal{H}} = e^{+is},$$

and in particular

$$\langle H_s, \Phi_- \rangle_{\mathcal{H}} = \overline{\langle H_s, \Phi_+ \rangle_{\mathcal{H}}}.$$

Proof. Using the isometry V of Theorem 7 and the definition $\Phi_{\pm} = V^{-1}(\mathbf{1}_{\{\pm 1\}})$,

$$\langle H_s, \Phi_- \rangle_{\mathcal{H}} = \langle V H_s, V \Phi_- \rangle_{L^2(\nu)} = \langle e^{i\xi s}, \mathbf{1}_{\{-1\}} \rangle_{L^2(\nu)} = \int_{\mathbb{R}} e^{i\xi s} \mathbf{1}_{\{-1\}}(\xi) d\nu(\xi) = e^{-is}.$$

Similarly $\langle H_s, \Phi_+ \rangle_{\mathcal{H}} = e^{+is}$, and $\overline{e^{+is}} = e^{-is}$. □

5 The spectral representation of the Hardy Z -function

Recall $Z(t) = \sqrt{\theta'(t)} H(\theta(t))$ on $[T_0, \infty)$. Under the coordinate change $x = \theta(t)$, the shift T_s on the x -axis becomes a pullback along θ ; in $\mathcal{H}$ this is expressed simply as substitution.

Theorem 14 (Spectral representation of the Hardy Z -function). *For every $t \geq T_0$, define $Z_t := \sqrt{\theta'(t)} \cdot H_{\theta(t)} \in \mathcal{H}$, where $H_{\theta(t)} := V^{-1}(e^{i\xi\theta(t)})$. Then*

$$Z_t = \sqrt{\theta'(t)} \int_{\mathbb{R}} e^{i\xi\theta(t)} d\Phi(\xi) = \sqrt{\theta'(t)} (e^{-i\theta(t)} \Phi_- + e^{+i\theta(t)} \Phi_+) \quad \text{in } \mathcal{H}. \quad (10)$$

Moreover, the limit covariance $\mathcal{C}_Z(t, s) := \langle Z_t, Z_s \rangle_{\mathcal{H}}$ is

$$\mathcal{C}_Z(t, s) = \sqrt{\theta'(t)\theta'(s)} \cdot C(\theta(t) - \theta(s)) = 2\sqrt{\theta'(t)\theta'(s)} \cos(\theta(t) - \theta(s)).$$

Proof. The defining identity $Z_t = \sqrt{\theta'(t)} H_{\theta(t)}$ is an equation in $\mathcal{H}$ by scalar multiplication and the element $H_{\theta(t)} = V^{-1}(e^{i\xi\theta(t)}) \in \mathcal{H}_H$ is well-defined because $\theta(t) \in \mathbb{R}$ for every t and the exponential is in $L^2(\nu)$. Substituting Theorem 11 applied at $x = \theta(t)$ and Corollary 12 yields the two forms in (10).

For the covariance, use bilinearity:

$$\begin{aligned} \mathcal{C}_Z(t, s) &= \langle \sqrt{\theta'(t)} H_{\theta(t)}, \sqrt{\theta'(s)} H_{\theta(s)} \rangle_{\mathcal{H}} \\ &= \sqrt{\theta'(t)\theta'(s)} \langle H_{\theta(t)}, H_{\theta(s)} \rangle_{\mathcal{H}} \\ &= \sqrt{\theta'(t)\theta'(s)} C(\theta(t) - \theta(s)) \end{aligned}$$

by (6), and $C(\theta(t) - \theta(s)) = 2 \cos(\theta(t) - \theta(s))$ by Input 1. □

Corollary 15 (Diagonal variance). $\langle Z_t, Z_t \rangle_{\mathcal{H}} = 2\theta'(t)$ for every $t \geq T_0$.

Proof. Set $s = t$ in Theorem 14; $C(0) = 2$, $\theta(t) - \theta(t) = 0$. □

6 Remarks on terminology

Remark 16 (The word “random”). The object Φ constructed in Definition 8 is a finitely-additive, countably-additive-in-norm map from the Borel σ -algebra of $\mathbb{R}$ into the Hilbert space $\mathcal{H}$, with $\langle \Phi(A), \Phi(B) \rangle_{\mathcal{H}} = \nu(A \cap B)$. This is exactly the definition of a complex orthogonally-scattered vector measure on $\mathcal{B}(\mathbb{R})$ with intensity ν and values in $\mathcal{H}$, in the sense of Masani–Rosenberg. It is called the *complex orthogonal measure* of the signal H .

Every such vector-valued measure is isomorphic, via the Kolmogorov construction, to the complex orthogonal increment measure of a second-order centered process on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ whose covariance is $\nu(A \cap B)$: namely, take Ω to be a Gaussian measure space with reproducing kernel Hilbert space $\mathcal{H}$. Under that isomorphism $\mathcal{H} \ni f \mapsto \tilde{f} \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ with $\tilde{f}$ a complex Gaussian and $\mathbb{E}[\tilde{f}\tilde{g}] = \langle f, g \rangle_{\mathcal{H}}$, the map Φ becomes a genuine complex Gaussian orthogonal random measure; this is the sense of “random” used in the companion literature. The present paper does not require this extra step: the identities (8) and (10) hold in $\mathcal{H}$ without invoking any probability space.

Remark 17 (Two atoms is the content). The support of ν is $\{-1, +1\}$ and $\mathcal{H}_H$ is two-dimensional; under V , $\mathcal{H}_H \simeq L^2(\nu) \simeq \mathbb{C}^2$ via $f \mapsto (Vf(-1), Vf(+1))$. In particular $\{H_s : s \in \mathbb{R}\}$ spans only a two-dimensional subspace of $\mathcal{H}$: the limit structure of H records only the phase pair (± 1) . This is consistent with the band-edge calculation of the companion paper *Existence of the Spectral Measure in the Limit*, where the same two atoms appeared as the Bochner measure of $C(\eta) = 2 \cos |\eta|$.

Remark 18 (Relation to the pre-limit harmonizable representation). At finite X , the pre-limit measure ν_X is absolutely continuous with density $|\widehat{H_X}|^2/(X - X_0)$ where $\widehat{H_X}(\xi) = \int_{X_0}^X H(x)e^{-i\xi x} dx$; see *Existence of the Spectral Measure in the Limit*, Proposition 9. By Theorem 9, the complex orthogonal measure Φ constructed here is the limit-in- $\mathcal{H}$ of the pre-limit complex orthogonal measures Φ_X obtained by applying the same Kolmogorov isometry to ν_X in place of ν : the isometries $V_X: \mathcal{H}_{H,X} \rightarrow L^2(\nu_X)$ exist at every X (for every positive finite Borel measure ν_X there is such an isometry sending $H_s \mapsto e^{i\xi s}$), and under the distributional convergence $\nu_X \rightarrow \nu$ in $\mathcal{S}'(\mathbb{R})$ their transported images in a common Hilbert space converge to Φ . The two-atom limit $\nu = \delta_{-1} + \delta_{+1}$ then concentrates the pre-limit continuous densities onto the atoms $\Phi_{\pm}$.

Summary of proved statements

- **Hilbert space exists** (Definitions 1–4, Lemmas 2–3): the shifts $T_s H$ of the warped Hardy signal lie in a Hilbert space $\mathcal{H}$ where the inner product is the time-average covariance $\langle H_s, H_t \rangle = C(s - t) = 2 \cos(s - t)$, with C from Input 1.
- **Complex orthogonal measure exists** (Theorem 7, Definition 8, Theorem 9): there is a unique Hilbert-space-valued orthogonally-scattered measure Φ on $\mathcal{B}(\mathbb{R})$ with intensity $\nu = \delta_{-1} + \delta_{+1}$, supported on $\{-1, +1\}$, whose two atoms $\Phi_{\pm}$ are orthonormal in $\mathcal{H}$.
- **Spectral representation of H** (Theorem 11, Corollary 12): $H_x = \int e^{i\xi x} d\Phi(\xi) = e^{-ix} \Phi_- + e^{+ix} \Phi_+$ in $\mathcal{H}$.
- **Spectral representation of Z** (Theorem 14): $Z_t = \sqrt{\theta'(t)} \int e^{i\xi \theta(t)} d\Phi(\xi) = \sqrt{\theta'(t)} (e^{-i\theta(t)} \Phi_- + e^{+i\theta(t)} \Phi_+)$ in $\mathcal{H}$, with covariance $\langle Z_t, Z_s \rangle_{\mathcal{H}} = 2\sqrt{\theta'(t)\theta'(s)} \cos(\theta(t) - \theta(s))$.